Engineering Maintainance Intern

FINAL INTERNSHIP REPORT

Submitted by

Harsh Raj Jaiswal 21BAS1236

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

AEROSPACE ENGINEERING

Chandigarh University

Mohali, Punjab

April 2025

Offer letter

NEPAL AIRLINES CORPORATION
Human Resources Department
KATHMANDU

Nepal Airlines

Ref.No. HRD/18/1169 /2081-082

Date: 2081/09/14
2024/12/29

Mr. Harsh Raj Jaiswal
Kaudena -03, Sarlahi

Subject: Internship

Dear, Mr. Jaiswal

This refers to your application dated 2081/08/23 regarding the internship in our airlines.

In this regard, we would like to inform you that your request to work as 'Intern' in Engineering Maintenance Department has been accepted under the following terms and conditions:

1. That you have been placed as internee for six (06) months with effect from your date of attendance in the Engineering Maintenance Department.
2. That you should be in college uniform with the corporation identity card and tie provided by Nepal Airlines Corporation during the working office hour and maintain official decorum.
3. Nepal Airlines will not bear any financial burden during your internee period.
4. That you will be solely responsible for your indulgence in any illegal acts and/ or acts being forbidden by law.
5. You will be made liable of financial loss, if any, incurred by your negligence.
6. That your internship will be discontinue if fail to abide by instructions, rules & regulations of NAC.
7. Nepal Airlines will process Airport Pass for you from Civil Aviation Authority of Nepal (CAAN) and charge for the same will be borne by yourself.
8. You will be charged your internship fee in advance at the rate of Rs. 6000.00 (Rupees Six Thousand Only) per month.
9. After completion of your internee period, you will be provided a certificate accordingly.

You will be under supervision of Director; Engineering Maintenance Department and will be provided with letter of placement with specific responisbilities.

Please sign on appropriate space below as your acceptance on above terms and conditions.

Ashok Sigdel
Actg. Director
Human Resources Department

Accepted by
Harsh Raj Jaiswal

BONAFIDE CERTIFICATE

This is to Certify that **Harsh Raj Jaiswal (In 1026)** has been doing internship from January 2025 under the guidance of Director of Engineering Maintenance Department, Nepal Airline Corporation.

Mr. Pawan Kumar Karwal
Director/Maintenance Manager
Engineering Maintenance Department
Nepal Airline Corporation
Kathmandu, Nepal

The reports of the project work submitted by the above student in partial fulfillment for the award of Bachelor of Engineering degree In Aerospace Engineering of Chandigarh University were evaluated and confirmed to be the reports of the work done by the above students and then evaluated.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGMENT

I would like to express my sincere gratitude to all those who supported and guided me during my six-month internship as an **Engineering Maintenance Intern** at **Nepal Airlines Corporation**, Kathmandu.

First and foremost, I am deeply thankful to **Pawan Kumar Karwal**, **Director of the Engineering Maintenance Department at Nepal Airlines Corporation, Kathmandu, Nepal**, for granting me the opportunity to be a part of the maintenance team and for allowing me to gain practical exposure to aircraft systems and operations. This opportunity has significantly enriched my technical knowledge and understanding of real-world aviation maintenance.

I would also like to extend my appreciation to **Gopal Prasad Tiwari**, **EMD Administrative Head**, for their assistance and support in managing the necessary documentation and coordination related to the internship process.

I am equally grateful to **Prof. Dharmahinder Singh Chand**, **Head of Department, Aerospace Engineering, Chandigarh University**, for permitting and encouraging me to pursue this internship as part of my academic curriculum. Their support played an essential role in making this experience possible.

This internship has been a valuable learning experience, and I am thankful to all those who contributed to its success.

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Abbreviations

AME	Aircraft Maintenance Engineer
AMM	Aircraft Maintenance Manual
APU	Auxiliary Power Unit
ATA	Air Transport Association (Chapter Code System)
AD	Airworthiness Directive
ADIRS	Air Data Inertial Reference System
ATC	Air Traffic Control
BITE	Built-In Test Equipment
CAAN	Civil Aviation Authority of Nepal
DFDR	Digital Flight Data Recorder
ECAM	Electronic Centralized Aircraft Monitor
ECS	Environmental Control System
EFCS	Electrical Flight Control System
FOD	Foreign Object Damage
FIM	Fault Isolation Manual
GVI	General Visual Inspection
IFE	In-Flight Entertainment
IPC	Illustrated Parts Catalogue
LRU	Line Replaceable Unit
MEL	Minimum Equipment List
MCC	Maintenance Control Center
MLG	Main Landing Gear
MRO	Maintenance, Repair, and Overhaul
NAC	Nepal Airlines Corporation
OEM	Original Equipment Manufacturer
PPE	Personal Protective Equipment
RAFT	Remote Area Firefighting Tool (or inflatable life raft)
SOP	Standard Operating Procedure
STOL	Short Take-Off and Landing
THSA	Trimmable Horizontal Stabilizer Actuator
T3CAS	Terrain, Traffic, and Turbulence Collision Avoidance System
V.I.	Visual Inspection

Chapter 1: Introduction

As a part of the academic curriculum for the Bachelor of Technology in Aerospace Engineering at Chandigarh University, every student is required to undergo an industrial internship to gain practical exposure and hands-on experience in the aviation and aerospace industry. In fulfilment of this requirement, I undertook a six-month internship as an Engineering Maintenance Intern at Nepal Airlines Corporation, Kathmandu, starting from January 1st, 2025.

This internship provided me with a valuable opportunity to apply theoretical knowledge acquired during the course to real-world aircraft maintenance operations. Nepal Airlines Corporation, being the national flag carrier of Nepal, operates a fleet consisting of both narrow-body and wide-body aircraft, including Airbus A320 and A330 series. The organization's Engineering and Maintenance Division plays a critical role in ensuring flight safety, reliability, and airworthiness of its fleet through rigorous inspection and servicing protocols.

During my tenure, I was exposed to various aspects of aircraft maintenance including routine checks, troubleshooting, component replacements, technical documentation, and compliance with international safety and maintenance standards. This internship has significantly contributed to my understanding of the practical challenges and responsibilities involved in aircraft maintenance and has further strengthened my passion for the aviation industry.

This report presents a detailed overview of the work undertaken during the internship, the skills acquired, and the insights gained throughout the training period.

Figure 1: Picture of Nepal Airline Maintenance Department

1.1 Objective of Internship

The primary objective of this internship was to gain hands-on experience in the field of **aviation maintenance** with a specific focus on **aircraft systems**, **maintenance procedures**, and **regulatory compliance**. The key objectives were:

1. **Exposure to Real-world Aircraft Maintenance**: To observe and assist in the maintenance and servicing of different aircraft types, including the **Airbus A320**, **Airbus A330**, and **DHC-6 Twin Otter**, with an emphasis on **line** and **base maintenance operations**.

2. **Understanding Aircraft Systems**: To gain practical knowledge of aircraft systems such as **hydraulics**, **fuel systems**, **electrical systems**, and **avionics**, and understand how they are monitored and serviced.

3. **Development of Technical Skills**: To develop proficiency in using maintenance tools, diagnosing faults, replacing components, and applying **safety protocols** in a real-world setting.

4. **Learning Maintenance Documentation and Compliance**: To understand the importance of **maintenance logs**, **technical manuals**, and compliance with **Service Bulletins (SBs)**, **Airworthiness Directives (ADs)**, and other regulatory requirements.

5. **Problem-solving and Troubleshooting**: To actively engage in identifying and troubleshooting mechanical or electrical issues and assisting in developing efficient solutions.

6. **Exposure to the Work Environment**: To experience the challenges and requirements of working in a **dynamic aviation environment**, particularly within the unique operational setting of **Nepal Airlines Corporation**.

Through these objectives, the internship aimed to bridge the gap between academic knowledge and practical application, enhancing my understanding of the aviation industry and preparing me for a future career in aerospace engineering.

1.2 Duration and Location of the Internship

The internship was conducted from **January 1, 2025** to **June 30, 2025**, spanning a total of **six months**. During this period, I had the opportunity to work at **Nepal Airlines Corporation**, located in **Kathmandu**, Nepal, specifically at **Tribhuvan International Airport**.

The location provided a dynamic and practical learning environment as I was able to engage with aircraft operations, maintenance activities, and the daily challenges of working with a diverse fleet of aircraft. The **Maintenance Hangar** and **Field Operations** areas of the airport offered direct exposure to real-time aircraft servicing and maintenance tasks.

Figure 2: Picture of NAC Hanger

Chapter 2: About the Organization

2.1 Nepal Airline Corporation

Established in 1958, **Nepal Airlines Corporation (NAC)** is the **national flag carrier of Nepal** and a key pillar of the country's civil aviation sector. Headquartered in Kathmandu, the airline operates both **domestic and international routes**, connecting remote regions of Nepal as well as major cities in Asia and the Middle East. Its fleet includes modern jetliners such as the **Airbus A320 and A330**, and utility aircraft like the **DHC-6 Twin Otter**, which are essential for serving Nepal's mountainous terrain and short runways.

As an **Engineering Maintenance Intern**, I was privileged to gain hands-on exposure to the airline's **Line and Base Maintenance operations** at Tribhuvan International Airport. Working within NAC's well-structured **Engineering and Maintenance Division**, I observed and assisted in critical aircraft servicing tasks, gaining insight into real-time aviation maintenance practices and regulatory compliance procedures. The organization's dynamic environment and commitment to safety and operational excellence provided an ideal learning platform for developing practical skills aligned with my aerospace engineering education.

2.2 Maintenance Department Overview

The **Maintenance Department** at **Nepal Airlines Corporation** (NAC) plays a critical role in ensuring the safety, airworthiness, and operational readiness of the airline's fleet. The department is responsible for performing routine **line maintenance**, periodic **base maintenance**, and resolving any unplanned technical issues that may arise during operations. The Maintenance Department is equipped with a skilled team of engineers, technicians, and support staff who work in accordance with stringent safety and regulatory standards.

The department operates under the guidelines set by the **Civil Aviation Authority of Nepal (CAAN)**, ensuring that all maintenance procedures meet both national and international aviation safety standards. It is divided into two key areas of operation:

1. **Line Maintenance**: This refers to the maintenance activities performed on aircraft that are in service, typically carried out between flights or overnight at the airport. It includes:

    - **Pre-flight and post-flight inspections** of aircraft systems.
    - **Basic troubleshooting** of any technical issues.
    - **Servicing of aircraft components** such as engines, landing gear, and avionics.
    - **Fluid checks** (e.g., fuel, hydraulic fluids, oil) and replenishment.
2. **Base Maintenance**: This includes more in-depth, scheduled maintenance, typically performed during aircraft downtime when it is out of service for longer durations. The tasks involve:
    - **Heavy inspections** (such as C-checks and D-checks).
    - **Replacement of major components** and parts.
    - **Structural repairs** and modifications.
    - Detailed inspections of **engines**, **landing gear**, and **flight controls**.

The Maintenance Department also integrates a system of **maintenance documentation**, where each procedure and activity are logged in the aircraft's maintenance records. This includes tracking of service intervals, repairs, part replacements, and compliance with **Service Bulletins (SBs)** and **Airworthiness Directives (ADs)**.

The department utilizes advanced **diagnostic tools** and **specialized equipment** for servicing aircraft, ensuring that the fleet remains in optimal operating condition at all times.

During my internship, I had the opportunity to observe and assist in both **line** and **base maintenance** tasks, gaining exposure to various types of maintenance practices on aircraft such as the **Airbus A320**, **Airbus A330**, and **DHC-6 Twin Otter**.

2.3 Aircraft Overview

During my internship at **Nepal Airlines Corporation**, I had the opportunity to work with a diverse fleet that includes the **Airbus A320**, **Airbus A330**, and the **DHC-6 Twin Otter**. Each aircraft type has unique design characteristics, operational purposes, and maintenance requirements.

A. Airbus A320

The Airbus A320 is a narrow-body, short-to-medium-haul commercial jet that is widely used for both domestic and regional international operations. It is known for its fuel efficiency and advanced avionics.

- **Type:** Narrow-body, twin-engine jet

- **Engines:** CFM56-5B or IAE V2500
- **Passenger Capacity:** ~150–180
- **Range:** Approx. 6,100 km
- **Notable Features:**
    - Fly-by-wire flight control system
    - Electronic Centralized Aircraft Monitor (ECAM)
    - Triple hydraulic systems and advanced avionics

Figure 3: Picture of NAC's A320

B. Airbus A330

The Airbus A330 is a wide-body, twin-engine aircraft used primarily for long-haul international routes. It shares cockpit commonality with the A320 but differs greatly in size, range, and systems complexity.

- **Type:** Wide-body, long-range jet
- **Engines:** Rolls-Royce Trent 700 or GE CF6
- **Passenger Capacity:** ~250–300
- **Range:** Approx. 13,450 km
- **Notable Features:**
    - Dual Air Cycle Machines (ECS)
    - Common Type Rating with A320 for pilots
    - Extensive cargo systems and larger fuel capacity

Figure 4: Picture of NAC's Airbus A330

C. DHC-6 Twin Otter

The Twin Otter is a rugged, high-wing STOL (Short Take-Off and Landing) aircraft ideal for Nepal's mountainous terrain and short airstrips.

- **Type:** High-wing, turboprop utility aircraft
- **Engines:** Pratt & Whitney PT6A
- **Passenger Capacity:** ~19
- **Range:** Approx. 1,400 km
- **Key Characteristics:**
  - Excellent short-field performance
  - Simple mechanical and avionics systems
  - Operates reliably in remote and harsh environments

Figure 5: Picture of NAC's Twin Otter

2.4 Internship Roles and Responsibilities

As an **Engineering Maintenance Intern**, I worked under licensed aircraft maintenance engineers and technicians across line and base maintenance divisions. My responsibilities varied depending on the aircraft type and maintenance schedule.

A. Airbus A320

Working with the A320 fleet involved exposure to modern maintenance procedures and digital systems. My roles included:

- Assisting during **A-checks** and routine maintenance operations
- Supporting **component removal and replacement** such as:
  - Hydraulic filters
  - Fuel pumps
  - Batteries and sensors
- Observing **fly-by-wire control surface tests**
- Assisting in inspection of:
  - **Brake wear indicators**
  - **Landing gear assemblies**
  - **Hydraulic reservoirs**
- Learning the use of digital maintenance manuals including AMM (Aircraft Maintenance Manual) and FIM (Fault Isolation Manual)
- Documenting work done in **task cards and technical logs**

B. Airbus A330

The A330 offered an opportunity to understand wide-body aircraft systems. I was involved in:

- Participating in **longer maintenance schedules** involving deep inspections
- Observing **APU start-up procedures**, **bleed air system tests**, and **pressurization systems**
- Supporting inspection and servicing of:
  - **Oxygen systems**
  - **Fuel quantity indication system**
  - **Electrical power generation and distribution**

- Learning about **maintenance planning** and scheduling through coordination with the Maintenance Control Center (MCC)
- Exposure to **Service Bulletins (SBs)** and **Airworthiness Directives (ADs)** compliance

C. DHC-6 Twin Otter

The Twin Otter experience was more hands-on, especially useful for understanding older, manually-controlled systems:

- Performing **pre-flight and post-flight inspections**
- Supporting basic servicing tasks:
    - **Oil level checks**
    - **Propeller condition monitoring**
    - **Manual control surface deflection checks**
- Assisting in **field maintenance** operations at remote airports
- Verifying emergency and survival equipment such as:
    - Fire extinguishers
    - First aid kits
    - Seatbelt condition and oxygen masks
- Gaining practical insight into **rugged aircraft operations** in real-world conditions.

Chapter 3: Weekly Report

3.1 First Week on Internship

Table 1: First Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330-243	9N-ALY	• Check Torque of IDG quick detach tension bolt (ENG #2) • Drain IDG oil scavenge filter element n replenish (ENG#2) • Compliance of SB A330-25-3768, R03	
2		9N- ALZ	• Open check of network reconfiguration in mono generation configuration • Starter oil replenishment (ENG #1) • Disinsection of the A/C equipment and furnishing	
3	Airbus A320-233	9N-AKW	• Check wear indicator of starter motor brush • Remove door escape slide/RART • Download and readout the DFDR data	
4		9N-AKX	• Remove the thermostat-filter of the fan air valve control on zone 413 for cleaning • Replacement of R/H Nav-housing bolts • Operational check of PAX/crew door escape slide/RART deployment	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Engine preparation before installation	

3.2 Second Week on Internship

Table 2: Second Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Compliance of SB A330-25-3768, R03 • V.I of bush for signs of rotation fitting outer flange • Operational check of emergency generation system	
2		9N- ALZ	• Lubrication of MLG bogie beam Pivot • Function check of normal and alternate brake returns accumulators N2 Pressure by reading gauges • First aid kit items replacement	
3	Airbus A320	9N-AKW	• Remove, Inspect and Replace with a new or cleaned pressure filter on ENG#2 • Clean Chilled air in and chilled air out grills • Clean air chiller condenser air filter	
4		9N-AKX	• Clean filter element of wash basin drains valves	

			• Drain recovery tank (main hydraulic power) • Upload Navigation Database	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Engine preparation before installation	

3.3 Third Week on Internship

Table 3: Third Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Open check of emergency power supply units by BITE-check • Open check of lights and emergency generation system • R/I of Ni-Cd Battery 3(APU0	
2		9N- ALZ	• Cleaning and functional check of EMCD (ENG #1) • Cleaning of pivot probe on standby position • Operational check of FCPCS & FCSCS standby servo loops on pitch control • Visual inspection of bush for signs of rotation	
3	Airbus A320	9N-AKW	• Cleaning of pitot probe number 1(Pin:90A1) • G.V.I of inner barrel heatshield and fire seals • G.V.I of inner surface of doors and seals	
4		9N-AKX	• Check master MCD for evidence of metallic debris • Check of RH MLG shock absorber charging pressure • Discard IDG scavenge and charging filter	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Engine preparation before installation	

3.4 Fourth Week on Internship

Table 4: Fourth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Open check of THS actuator with individual hydraulic systems • Operate water drain valves to drain accumulated water from trim tank and trim vent surge tank • Cleaning of the carpets and NTF floor covering • Disinfection of potable water system	
2		9N- ALZ	• Discard water filter cartridge of the galleys	

			• LP compressor front case assembly local repair of the rear acoustic panels as per ADD B025542 • Starter oil replenishment (ENG #2)	
3	Airbus A320	9N-AKW	• Check wear indicator of starter motor brush • Operational check of PAX/crew door escape slide/RART deployment • Remove door escape slide/RART	
4		9N-AKX	• Remove, Inspect and Replace with a new or cleaned pressure filter on ENG#2 • Upload Navigation Database • Cleaning of pitot probe number 1(Pin:90A1)	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Battery installation R/I of RC-Relay • Engine dress up and propeller assembling	

3.5 Fifth Week on Internship

Table 5: Fifth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Placement of AED (Automated External Defibrillator) • Upload NAV Database • Discard megaphone batteries position forward/aft	
2		9N- ALZ	• Open check of DC battery bus and DC ESS buss isolation • Open check of static inverter and DC ESS bus supply • Aircraft total time correction	
3	Airbus A320	9N-AKW	• G.V.I of inner barrel heatshield and fire seals • G.V.I of inner surface of doors and seals • Check master MCD for evidence of metallic debris	
4		9N-AKX	• Check of RH MLG shock absorber charging pressure • Discard IDG scavenge and charging filter • Drain and replenish oil system on ENG#2	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Battery installation and engine fire bottle installation	

3.6 Sixth Week on Internship

Table 6: Sixth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Discard cabin temperature sensor filters • Starter oil change and inspection of starter chip detector (ENG #1 &2) • R/I of battery of ASPSU for in-shop battery capacity check • Remove, discard and replace oil scavenge filter (ENG 1&2)	
2		9N- ALZ	• Oxygen cylinder for in shop hydrostatic test and functional check of oxygen cylinder pressure gauge • Two impacts of dimension (3cm/3cm) on infill panel of outlet guide vane in RH engine • Replacement of affected parts of AET bulkhead cover panel in galley • Engine bleed air system main task	
3	Airbus A320	9N-AKW	• Clean filter element of wash basin drains valves • Upload IFE content and T3CAS terrain database • Operational check of light test system	
4		9N-AKX	• G.V.I of fan cowl • Disinsection of aircraft equipment and furnishings • First aid kit visual inspection	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Preparation of aircraft for engine installation	

3.7 Seventh Week on Internship

Table 7: Seventh Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• FAK expiry items replacement • Operational check of manual fire extinguishing circuit in the LDMCR • Removal and Installation of ULB for in shop ULB battery discard	
2		9N- ALZ	• Lubrication of MLG and doors • R/I of EPSU batteries • Replacement of dent and buckle chart	
3	Airbus A320	9N-AKW	• Borescope inspection of inside of combustion chamber and HP turbine stage 1 vanes on ENG#2	

			• Borescope inspection of HP turbine blade aerofoils on ENG#2 • Check of electrical flight control system (EFCS)	
4		9N-AKX	• Check of RH MLG shock absorber charging pressure • Discard IDG scavenge and charging filter • Seat Inspection • Operational check of rudder servo controls with individual hydraulic systems	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Preparation of aircraft for engine installation	

3.8 Eight Week on Internship

Table 8: Eight Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Cleaning and functional check of EMCD (ENG 1&2) • Operational check of SQUIB functionality in the LDMCR • Remove, discard and replace oil scavenge filter (ENG 1&2)	
2		9N- ALZ	• Oxygen cylinder for in shop hydrostatic test and functional check of oxygen cylinder pressure gauge • Two impacts of dimension (3cm/3cm) on infill panel of outlet guide vane in RH engine • Open check of static inverter and DC ESS bus supply	
3	Airbus A320	9N-AKW	• Operational check of rudder servo controls with individual hydraulic systems • General visual inspection of L/G brake units • Discard and replace megaphone batteries • Detailed inspection of the inboard MLG support rib lower flange to detect broken fastener tails/nuts	
4		9N-AKX	• Mapping of portable fire extinguisher • Mapping of portable oxygen cylinders and smoke hoods • Mapping of full-face donning mask • Cleaning of pilot probe no.3 (9DA3)	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Propeller assembling	

3.9 Nineth Week on Internship

Table 9: Nineth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Compliance of SB A330-25-3768, R03 • V.I of bush for signs of rotation fitting outer flange • Operational check of emergency generation system	
2		9N- ALZ	• Lubrication of MLG bogie beam Pivot • Function check of normal and alternate brake returns accumulators N2 Pressure by reading gauges • First aid kit items replacement	
3	Airbus A320	9N-AKW	• General visual inspection of L/G brake units • Discard and replace megaphone batteries • Detailed inspection of the inboard MLG support rib lower flange to detect broken fastener tails/nuts • Replace EPSU battery from fin 20WL	
4		9N-AKX	• Greasing of MLG door actuator • Position sensor wheel and optic sensor cleaning • Detailed inspection of the external surface of the LH and RH lower skin surface to detect missing or migrating fasteners • Replace EPSU battery from fin 20WL	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Propeller assembling	

3.10 Tenth Week on Internship

Table 10: Tenth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Cleaning and functional check of EMCD (ENG #1) • Cleaning of pivot probe on standby position • Operational check of FCPCS & FCSCS standby servo loops on pitch control • Visual inspection of bush for signs of rotation	
2		9N- ALZ	• Placement of AED (Automated External Defibrillator) • Upload NAV Database • Discard megaphone batteries position	

3	Airbus A320	9N-AKW	• Operational check of cargo component smoke detection by pit • Check pressure gauge reading • Replacement of ACOC modulating valve • General visual inspection of fuel tanks for minor external leaks • Operational check of water drains valves	
4		9N-AKX	• Repair of the weather seal of windshield • Removal of survival emergency locator and transmitter locator • Replacement of pack flow control valve and pack flow control unit • Parking periodic check for 7days interval • Microbiological contamination analysis of fuel sample	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Defect rectification of reported defect of booster pump	

3.11 Eleventh Week on Internship

Table 11: Eleventh Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Open check of emergency power supply units by BITE-check • Open check of lights and emergency generation system • R/I of Ni-Cd Battery 3(APU0	
2		9N- ALZ	• Cleaning and functional check of EMCD (ENG #1) • Cleaning of pivot probe on standby position • Operational check of FCPCS & FCSCS standby servo loops on pitch control • Visual inspection of bush for signs of rotation	
3	Airbus A320	9N-AKW	• Disinfection of potable water system and discard of water filter elements • Detailed inspection of the THSA upper attachment for primary and secondary load path • Detailed inspection of the inboard main landing gear support rib lower flange • G.V.I of galley 5 • Mapping of portable equipment	

4		9N-AKX	• Lubrication of gear unlock • Detailed inspection of all cockpit windows from outside including seals • Remove door escape slide / RAFT for in shop restoration	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Defect rectification of reported defect of booster pump	

3.12 Twelfth Week on Internship

Table 12: Twelfth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Check Torque of IDG quick detach tension bolt (ENG #2) • Drain IDG oil scavenge filter element n replenish (ENG#2) • Compliance of SB A330-25-3768, R03	
2		9N- ALZ	• Open check of network reconfiguration in mono generation configuration • Starter oil replenishment (ENG #1) • Disinsection of the A/C equipment and furnishing	
3	Airbus A320	9N-AKW	• Greasing of MLG door actuator part no.114122014 • G.V.I of inner barrel heatshield and fire seals • Check of RH MLG shock absorber charging pressure	
4		9N-AKX	• Borescope inspection of inside of combustion chamber and HP turbine stage 1 vanes on ENG#2 • Borescope inspection of HP turbine blade aerofoils on ENG#2 • Check of electrical flight control system (EFCS)	
5	DHC-6 Twin Otter	9N-ABU	• 3M inspection	
6		9N-ABT	• Defect rectification of reported defect of booster pump	

3.13 Thirteenth Week on Internship

Table 13: Thirteenth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Operate water drain valves to drain accumulated water from trim tank and trim vent surge tank • Cleaning of the carpets and NTF floor covering • Disinfection of potable water system	
2		9N- ALZ	• Discard water filter cartridge of the galleys • LP compressor front case assembly local repair of the rear acoustic panels as per ADD B025542 • Starter oil replenishment (ENG #2)	
3	Airbus A320	9N-AKW	• Check wear indicator of starter motor brush • Operational check of PAX/crew door escape slide/RART deployment • Remove door escape slide/RART	
4		9N-AKX	• Remove, Inspect and Replace with a new or cleaned pressure filter on ENG#2 • Upload Navigation Database • Cleaning of pitot probe number 1(Pin:90A1)	
5	DHC-6 Twin Otter	9N-ABU	• Engine preparation before installation	
6		9N-ABT	• No schedule maintenance	

3.14 Fourteenth Week on Internship

Table 14: Fourteenth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Cleaning and functional check of EMCD (ENG 1&2) • Operational check of SQUIB functionality in the LDMCR • Remove, discard and replace oil scavenge filter (ENG 1&2)	
2		9N- ALZ	• Oxygen cylinder for in shop hydrostatic test and functional check of oxygen cylinder pressure gauge • Two impacts of dimension (3cm/3cm) on infill panel of outlet guide vane in RH engine	

			• Open check of static inverter and DC ESS bus supply	
3	Airbus A320	9N-AKW	• Operational check of rudder servo controls with individual hydraulic systems • General visual inspection of L/G brake units • Discard and replace megaphone batteries • Detailed inspection of the inboard MLG support rib lower flange to detect broken fastener tails/nuts	
4		9N-AKX	• Mapping of portable fire extinguisher • Mapping of portable oxygen cylinders and smoke hoods • Mapping of full-face donning mask • Cleaning of pilot probe no.3 (9DA3)	
5	DHC-6 Twin Otter	9N-ABU	• Battery installation R/I of RC-Relay	
6		9N-ABT	• No schedule maintenance	

3.15 Fifteenth Week on Internship

Table 15: Fifteenth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Open check of emergency power supply units by BITE-check • Open check of lights and emergency generation system • R/I of Ni-Cd Battery 3(APU0	
2		9N- ALZ	• Cleaning and functional check of EMCD (ENG #1) • Cleaning of pivot probe on standby position • Operational check of FCPCS & FCSCS standby servo loops on pitch control • Visual inspection of bush for signs of rotation	
3	Airbus A320	9N-AKW	• Cleaning of pitot probe number 1(Pin:90A1) • G.V.I of inner barrel heatshield and fire seals • G.V.I of inner surface of doors and seals	
4		9N-AKX	• Check master MCD for evidence of metallic debris • Check of RH MLG shock absorber charging pressure • Discard IDG scavenge and charging filter	
5	DHC-6 Twin Otter	9N-ABU	• Engine dress up and propeller assembling	
6		9N-ABT	• No schedule maintenance	

3.16 Sixteenth Week on Internship

Table 16:Sixteenth Week Report

S.no.	Aircraft Name	Reg. no.	Task Description	Remark
1	Airbus A330	9N-ALY	• Compliance of SB A330-25-3768, R03 • V.I of bush for signs of rotation fitting outer flange • Operational check of emergency generation system	
2		9N- ALZ	• Lubrication of MLG bogie beam Pivot • Function check of normal and alternate brake returns accumulators N2 Pressure by reading gauges • First aid kit items replacement	
3	Airbus A320	9N-AKW	• Clean filter element of wash basin drains valves • Upload IFE content and T3CAS terrain database • Operational check of light test system	
4		9N-AKX	• G.V.I of fan cowl • Disinsection of aircraft equipment and furnishings • First aid kit visual inspection	
5	DHC-6 Twin Otter	9N-ABU	• Battery installation and engine fire bottle installation	
6		9N-ABT	• No schedule maintenance	

Chapter 4: Skills and Knowledge Gained

4.1 Tools Used for Aircraft Maintenance Department

1. Torque Wrench

Used for tightening bolts and nuts to a specific torque value as per manufacturer's recommendation.

Purpose: Prevents over/under-tightening which can cause component failure.

Figure 6: Torque Wrench

2. Multimeter

Used for measuring voltage, current, and resistance in aircraft electrical systems.

Purpose: For electrical fault diagnosis.

Figure 7: Multimeter

3. Borescope

For inspecting engine internals like turbine blades without disassembly.

Purpose: Non-destructive inspection.

Figure 8: Borescope

4. Pitot-Static Tester

Simulates altitude and airspeed to check pitot-static systems.

Purpose: Ensures instrument accuracy.

Figure 9: Pitot-Static Tester

5. Safety Wire Twister (Lockwire Pliers)

Used to secure fasteners against loosening due to vibration.

Purpose: Maintains fastener integrity.

Figure 10: Safety Wire Twister

6. Hydraulic Mule

Provides hydraulic pressure for system tests on the ground.

Purpose: Tests landing gear and flight controls.

Figure 11: Hydraulic Mule

7. Tool Trolleys & Shadow Boards

Keeps tools organized and prevents FOD.

Purpose: Maintains tool accountability.

Figure 12: Tool Trolleys

8. Aircraft Jack and Tripod Stands

Used for elevating aircraft during inspections.

Purpose: Allows safe access to landing gear and underside.

Figure 13: Aircraft Jack

4.2 Technical Learnings

During my six-month internship at **Nepal Airlines Corporation**, I gained substantial hands-on experience and practical knowledge in the field of aircraft maintenance. This internship provided a bridge between the theoretical principles learned in the classroom and their application in real-world aviation operations. The key technical learnings are outlined below:

1. Understanding of Aircraft Systems

- Gained in-depth knowledge of the **aircraft hydraulic, fuel, electrical, pneumatic, and avionics systems**, especially for **Airbus A320**, **Airbus A330**, and **DHC-6 Twin Otter**.
- Understood the operation and maintenance of **fly-by-wire systems**, **environmental control systems (ECS)**, **auxiliary power units (APUs)**, and **landing gear mechanisms**.
- Learned how different systems interact and how integrated system monitoring (like ECAM on Airbus aircraft) supports safe operations.

2. Maintenance Procedures and Documentation

- Learned to perform and assist in **line maintenance** and **base maintenance** tasks, including:
    - **A-check inspections**
    - **Pre-flight and post-flight inspections**
    - **Component servicing and replacement**
- Developed familiarity with **Aircraft Maintenance Manual (AMM)**, **Fault Isolation Manual (FIM)**, **Illustrated Parts Catalogue (IPC)**, and other OEM publications.
- Gained experience in **maintenance documentation**, including completing task cards, log entries, and understanding service bulletins (SBs) and airworthiness directives (ADs).

3. Use of Specialized Tools and Equipment

- Became proficient in handling aviation-specific tools such as:
    - **Torque wrenches**
    - **Multi-meters**
    - **Safety wire pliers**
    - **Borescopes**

  - **Pitot-static testers**
  - **Hydraulic mules**
- Learned the importance of **tool calibration, handling safety, and tool accountability** to avoid FOD (Foreign Object Damage).

4. Troubleshooting and Diagnostics

- Observed and assisted in diagnosing **electrical faults**, **hydraulic leaks**, and **sensor malfunctions** in various aircraft systems.
- Participated in borescope inspections of engine components to detect signs of wear or damage.
- Learned to interpret cockpit fault messages and apply troubleshooting procedures using the FIM and electronic central maintenance systems.

5. Safety and Compliance Awareness

- Gained a clear understanding of **aviation safety regulations**, **standard operating procedures (SOPs)**, and **human factors in maintenance**.
- Understood the significance of **safety briefings**, **PPE usage**, and **aircraft ground handling precautions**.
- Learned about compliance with **CAAN**, **ICAO**, and **EASA** maintenance regulations and procedures.

6. Exposure to Digital Systems and Navigation Databases

- Assisted in **uploading navigation and IFE (In-Flight Entertainment) databases**.
- Became familiar with **Digital Flight Data Recorder (DFDR) downloads and readouts**.
- Understood how modern aircraft utilize **digital flight control systems (DFCS)** and **air data computers (ADCs)**.

7. Aircraft-Specific Knowledge

- Acquired practical knowledge of:
  - **Airbus A320/330 fly-by-wire systems**, ECAM interfaces, and modular LRUs
  - **Twin Otter mechanical systems** suited for rugged and remote operations

    - Differences in maintenance needs between **narrow-body**, **wide-body**, and **turboprop aircraft**

This technical learning experience has not only enhanced my core engineering competencies but also prepared me for future roles in the aviation and aerospace industries. It instilled in me the discipline, precision, and situational awareness essential for working in high-stakes technical environments.

4.3 Safety and Compliance

Safety and regulatory compliance are the foundation of all operations within the aviation maintenance sector. During my internship at **Nepal Airlines Corporation**, I gained a deep understanding of how **safety protocols** and **compliance standards** are integrated into every stage of aircraft maintenance. These learnings were essential in helping me develop a disciplined and professional approach to working in such a high-risk, precision-driven environment.

1. Workplace Safety Protocols

- I was trained to strictly follow **standard operating procedures (SOPs)** and **maintenance safety protocols**, including:
    - Use of **personal protective equipment (PPE)** like gloves, safety goggles, and ear protection
    - Observance of **tool control procedures** to avoid foreign object damage (FOD)
    - Proper handling of **hazardous materials** such as hydraulic fluid, jet fuel, and cleaning solvents
- Attended safety briefings and toolbox talks regularly, which reinforced the importance of **hazard identification**, **emergency response**, and **risk mitigation** in the hangar and ramp areas.

2. Compliance with Aviation Regulations

- I learned how maintenance work must adhere to both **national** and **international** regulations issued by:
    - **Civil Aviation Authority of Nepal (CAAN)**
    - **International Civil Aviation Organization (ICAO)**
    - **European Union Aviation Safety Agency (EASA)**

- Observed how aircraft maintenance activities are planned and documented in compliance with regulatory standards, particularly during **aircraft inspections**, **component replacements**, and **record-keeping**.

Figure 14: Collage of Safety and Hazard Identification

3. Aircraft Ground Safety

- I was made aware of strict protocols for operating around aircraft on the ground. These included:
  - **Chocking wheels**, **installing safety pins**, and setting **ground locks** before performing inspections
  - Observing **no-go zones** around running engines and rotating propellers
  - Ensuring **fire safety** near refuelling areas and during engine servicing

4. Human Factors in Maintenance

- Understood the significance of **human error** in aircraft maintenance and its potential impact on safety.
- Learned about the **"Dirty Dozen"** — 12 common human factors that can lead to maintenance errors, such as fatigue, lack of communication, distraction, or stress.
- Was encouraged to speak up and seek help if unsure, which promoted a **just culture** that prioritizes safety over hierarchy.

5. Documentation and Airworthiness

- Participated in filling **task cards**, **logbooks**, and **maintenance tracking systems** under supervision.
- Learned the importance of **traceability**, ensuring that every task performed is logged with appropriate signatures and part numbers for **audit and regulatory review**.
- Gained awareness of how **Airworthiness Directives (ADs)**, **Service Bulletins (SBs)**, and **Minimum Equipment Lists (MELs)** guide compliance actions.

6. Safety Audits and Inspections

- Observed internal audits and aircraft inspections conducted by the **Quality Assurance Team**.
- Understood how safety compliance is regularly verified to maintain both the **aircraft's airworthiness** and the **organization's regulatory approval**.

This exposure to safety and compliance practices has significantly enhanced my understanding of the aviation industry's **zero-tolerance approach** to risk and reinforced the idea that in aviation, **"safety is never optional."** It has shaped my mindset toward adopting a meticulous, responsible, and regulation-driven approach in my future career.

4.4 Soft Skills Development

In addition to technical knowledge and hands-on experience, my internship at **Nepal Airlines Corporation** played a vital role in enhancing my **soft skills**—a critical aspect of becoming a well-rounded aerospace professional. These skills are essential for effective communication, teamwork, adaptability, and problem-solving in the high-stakes aviation environment.

1. Communication Skills

- Regular interactions with engineers, technicians, supervisors, and administrative staff helped improve my **verbal and written communication**.
- Learned how to **ask clear technical questions**, **report observations**, and **document maintenance tasks** precisely and professionally.
- Understood the importance of **listening actively** and following instructions accurately in a safety-critical environment.

2. Teamwork and Collaboration

- Worked closely with multidisciplinary teams including avionics, airframe, and powerplant specialists.
- Learned how to coordinate tasks, **share responsibilities**, and **support senior staff** during maintenance operations.
- Developed respect for team roles and the importance of **collaborative decision-making**, especially during time-sensitive procedures.

3. Time Management

- Understood the importance of adhering to maintenance schedules, especially during quick aircraft turnarounds and daily checks.
- Prioritized tasks efficiently and learned to manage time without compromising on quality or safety.
- Developed the ability to plan ahead, ensuring that tools, documents, and spare parts were available before beginning tasks.

4. Adaptability and Flexibility

- Adapted to working in **different environments**—from well-equipped hangars to remote field maintenance for the **DHC-6 Twin Otter**.
- Handled changes in work plans due to weather conditions, aircraft availability, or unplanned technical snags.
- Remained calm and focused under pressure while observing or participating in critical maintenance operations.

5. Problem-Solving and Critical Thinking

- Learned to approach technical problems methodically—**identify the fault**, **refer to manuals**, **consult experienced staff**, and apply logical reasoning.
- Observed and participated in real-time troubleshooting and fault isolation on various aircraft systems.
- Developed a mindset of **continuous learning** and improvement, essential in solving dynamic engineering challenges.

6. Professionalism and Work Ethics

- Maintained punctuality, discipline, and a **safety-first attitude** throughout the internship.
- Understood the value of **confidentiality**, **integrity**, and **responsibility** in a high-risk industry.
- Actively participated in daily briefings and reviews, showing eagerness to learn and grow.

This development of soft skills, alongside technical competencies, has prepared me to be a better team player, an effective communicator, and a responsible aviation professional. These skills will be invaluable not only in future internships and jobs but also in leadership roles in the aerospace industry.

Chapter 5: Challenges Faced and Its Solution

During my internship at **Nepal Airlines Corporation**, I encountered several challenges that tested my adaptability, problem-solving abilities, and ability to work under pressure. These challenges not only helped me grow as a professional but also provided me with practical solutions that I can apply in future roles. Below are some of the key challenges I faced during my internship and the solutions or strategies I adopted to overcome them:

1. Learning the Complexities of Aircraft Systems

Challenge: One of the initial challenges I faced was the steep learning curve associated with understanding the various complex systems on the aircraft, such as the **hydraulic system**, **avionics**, and **fuel systems**. As a theoretical student, the complexity of real-world systems, especially on large aircraft like the **Airbus A330**, was daunting.

Solution: To overcome this, I dedicated extra time to study the **Aircraft Maintenance Manual (AMM)**, **Service Bulletins (SBs)**, and **Airworthiness Directives (ADs)** related to the aircraft I was working on. I also sought help from my mentors and senior technicians, who were generous in sharing their practical insights and experiences. Gradually, I became more comfortable with troubleshooting and identifying issues in these systems.

2. Time Pressure and Meeting Maintenance Deadlines

Challenge: During busy periods, such as aircraft turnaround after international flights, the team faced time constraints in completing routine maintenance checks. This often resulted in a high-pressure work environment where tasks had to be completed swiftly without compromising safety or quality.

Solution: To manage this, I learned the importance of time management and prioritization. I collaborated closely with senior engineers and technicians to develop an efficient workflow for routine checks. By breaking down tasks into smaller, manageable components and ensuring that no step was overlooked, we were able to complete inspections and repairs within the required timeframe. Additionally, I familiarized myself with the **checklist-based approach** for maintenance to ensure systematic progress and avoid mistakes due to haste.

3. Troubleshooting Unfamiliar Problems

Challenge: On several occasions, I was tasked with troubleshooting issues that were not familiar to me, such as **electrical faults** or **unresponsive avionics**. These problems required quick thinking and precise diagnosis, and I found myself unsure about which diagnostic tools to use or what steps to take.

Solution: In these instances, I relied on the guidance of experienced technicians and engineers. I also referred to the **maintenance manuals**, used **diagnostic equipment** to gather data, and consulted past maintenance logs for similar problems. Over time, I became more confident in my troubleshooting abilities. I also learned the importance of a systematic approach to diagnosing and fixing issues, which helped me avoid jumping to conclusions without fully understanding the problem.

4. Understanding Maintenance Documentation and Regulations

Challenge: Another challenge I faced was understanding the depth and importance of **maintenance documentation**. Aircraft maintenance requires strict adherence to records and regulatory standards. Initially, navigating through the extensive documentation and ensuring compliance with aviation regulations was overwhelming.

Solution: To overcome this, I took a proactive approach in learning the structure of **maintenance logbooks**, the significance of **task cards**, and the procedures for documenting inspections and repairs. I worked closely with administrative staff to understand how maintenance records were maintained and filed. I also learned about the regulatory environment, which included **Airworthiness Directives (ADs)**, **Safety of Life at Sea (SOLAS)**, and other compliance measures. With guidance and practice, I became more proficient in ensuring that all tasks were properly recorded and compliant with aviation standards.

5. Adapting to Field Maintenance in Remote Locations

Challenge: One of the more challenging aspects of my internship involved **field maintenance** operations, particularly when I was sent to remote locations for servicing the **DHC-6 Twin Otter** aircraft. These field conditions posed several logistical issues, such as limited access to tools, spare parts, and challenging weather conditions.

Solution: To adapt, I learned the importance of **preparation** and **planning**. Before traveling to a remote location, I ensured that all necessary tools and spare parts were carried along. I also gained a better understanding of the unique maintenance needs of **STOL (Short Take-off and Landing)** aircraft like the **Twin Otter**, which are often used in more rugged and inaccessible environments. Collaborating with the field team, I learned to perform basic repairs using available resources and prioritized essential tasks that would ensure the aircraft's operational safety upon return to base.

Chapter 6: Conclusion

In conclusion, my internship at **Nepal Airlines Corporation** provided an invaluable learning experience that greatly enhanced my theoretical knowledge in aerospace engineering. Over the course of six months, I had the opportunity to work alongside experienced professionals in the **Maintenance Department**, gaining hands-on experience with the **Airbus A320**, **Airbus A330**, and **DHC-6 Twin Otter** aircraft.

The exposure to both **line maintenance** and **base maintenance** operations allowed me to deepen my understanding of aircraft systems, safety protocols, and industry standards. I gained practical knowledge of maintenance activities such as **pre-flight inspections**, **hydraulic system servicing**, and **routine checks** of avionics, electrical systems, and landing gear. Furthermore, the real-world application of maintenance documentation and the importance of compliance with **Airworthiness Directives (ADs)** and **Service Bulletins (SBs)** has been a key takeaway from this internship.

This experience not only developed my technical expertise but also improved my **problem-solving skills**, **team collaboration**, and ability to work under pressure in a high-stakes environment. I learned how to troubleshoot technical issues, prioritize tasks, and contribute to ensuring the operational readiness of aircraft, which are critical to the airline's safety and efficiency.

I am deeply grateful to **Nepal Airlines Corporation** for offering me this internship and to all the professionals who mentored and guided me throughout. The skills and insights I gained will undoubtedly play a significant role in my future career in aerospace engineering. This internship has not only broadened my technical horizons but also reaffirmed my commitment to pursuing a career in the aviation industry.

Future Scope

The aviation and aerospace industry are undergoing rapid transformation, driven by technological innovation, increased regulatory focus on safety, and a growing emphasis on sustainability. My internship at **Nepal Airlines Corporation** has not only provided me with a solid foundation in aircraft maintenance but also inspired me to explore the broader potential of my career in this dynamic field. Based on my learning and observations, the following areas present significant future scope:

1. Advancements in Aircraft Maintenance Technology

- The increasing adoption of **predictive maintenance** using **Artificial Intelligence (AI)** and **Machine Learning (ML)** will revolutionize how maintenance tasks are planned and executed.

- Integration of **Internet of Things (IoT)** sensors in aircraft systems will allow for real-time health monitoring and more accurate fault detection.
- As an engineer, being proficient in **digital tools**, **data analytics**, and **automated diagnostic systems** will be essential to stay relevant in the industry.

2. Growing Demand for MRO Services

- The global **Maintenance, Repair, and Overhaul (MRO)** market is expanding rapidly, especially in Asia-Pacific regions, creating demand for skilled aircraft maintenance engineers.
- Airlines and MRO providers will increasingly seek professionals with cross-platform knowledge (Airbus, Boeing, and turboprop aircraft) and multi-skill certifications.
- There is an opportunity for career growth not only in airlines but also in **independent MRO firms**, **OEMs**, and **aerospace startups**.

3. Emphasis on Safety and Regulatory Compliance

- With aviation safety becoming more regulated, there is a growing scope for professionals in **compliance auditing**, **safety analysis**, and **quality assurance** roles.
- Opportunities also exist in government aviation bodies like **DGCA**, **CAAN**, **EASA**, and **ICAO**, where engineering insight is critical for rule-making and safety monitoring.

4. Green Aviation and Sustainability

- The aviation industry is moving toward greener technologies including **electric propulsion**, **sustainable aviation fuels (SAFs)**, and **hybrid aircraft**.
- Maintenance procedures will evolve to accommodate new systems, materials, and environmental standards, creating a new skill niche for eco-conscious engineers.

5. Continuous Professional Development

- To meet future challenges, I aim to pursue advanced certifications such as:
    - **EASA Part-66 AME License**
    - **DGCA CAR-66 License (India)**
    - Specialized training in **Avionics**, **NDT (Non-Destructive Testing)**, and **Borescope Inspection**

I also plan to further my education with a **Master’s in Aerospace Engineering or Aviation Safety**, which will enable me to contribute at a higher level in engineering design, systems integration, or regulatory strategy.

This internship has laid the groundwork for my professional journey in the aviation industry. With continuous learning, skill enhancement, and exposure to emerging technologies, I am confident that I can contribute meaningfully to the future of aerospace engineering and aircraft maintenance. The scope for growth is vast—not just in terms of career opportunities, but in shaping the safety, efficiency, and sustainability of the next generation of air travel.

Annexure

Figure 15: Picture of Tyre of A320

Figure 18: Picture of Twin Otter Engin without outer case

Figure 16: Picture of Technician correcting the meter readings

Figure 19: Picture of A330 Entry way connected with staircase

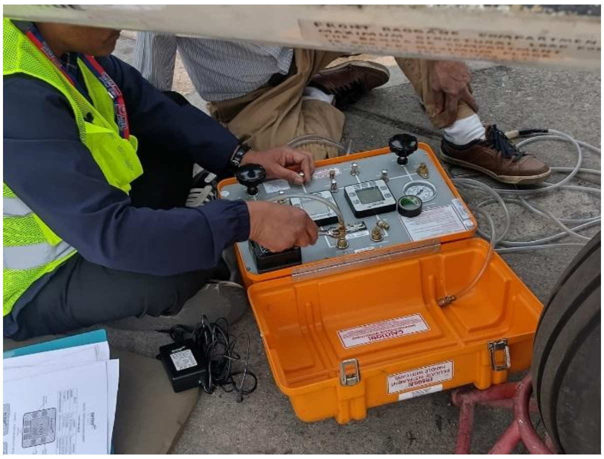

Figure 17: Picture of Technician using Pitot-static Tube

Figure 20: Picture of Backside of the A330 Engine

Figure 21: Picture of Cockpit of A330

Figure 22: Picture of Technician installing the engine

Figure 23: Picture of Ajay with A320 Tail

Figure 24: Picture of Harsh with A320 Tai